

1. COMMIT Program

`</> SQL`

```
CREATE DATABASE BankDB;
```

```
USE BankDB;
```

```
CREATE TABLE Account (  
    Account_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Balance INT  
);
```

```
INSERT INTO Account VALUES (101, 'Rahul', 10000);
```

```
INSERT INTO Account VALUES (102, 'Priya', 15000);
```

```
COMMIT;
```

```
SELECT * FROM Account;
```

COMMIT permanently saves the changes.

2. ROLLBACK Program

`</> SQL`

```
CREATE DATABASE BankDB2;
```

```
USE BankDB2;
```

```
CREATE TABLE Account (  
    Account_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Balance INT  
);
```

```
INSERT INTO Account VALUES (101, 'Rahul', 10000);
```

```
INSERT INTO Account VALUES (102, 'Priya', 15000);
```

```
COMMIT;
```

```
START TRANSACTION;
```

```
UPDATE Account  
SET Balance = Balance - 2000  
WHERE Account_ID = 101;
```

```
ROLLBACK;
```

```
SELECT * FROM Account;
```

ROLLBACK cancels the changes made after the transaction started.

3. SAVEPOINT Program

SQL

```
CREATE DATABASE BankDB3;
```

```
USE BankDB3;
```

```
CREATE TABLE Account (  
    Account_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Balance INT  
);
```

```
INSERT INTO Account VALUES (101, 'Rahul', 10000);
```

```
INSERT INTO Account VALUES (102, 'Priya', 15000);
```

```
COMMIT;
```

```
START TRANSACTION;
```

```
UPDATE Account  
SET Balance = Balance - 2000  
WHERE Account_ID = 101;
```

```
SAVEPOINT sp1;
```

```
UPDATE Account  
SET Balance = Balance + 2000  
WHERE Account_ID = 102;
```

```
COMMIT;
```



SAVEPOINT creates a point inside a transaction to which you can later roll back.

4. ROLLBACK TO SAVEPOINT Program

</> SQL

```
CREATE DATABASE BankDB4;
```

```
USE BankDB4;
```

```
CREATE TABLE Account (  
    Account_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Balance INT  
);
```

```
INSERT INTO Account VALUES (101, 'Rahul', 10000);
```

```
INSERT INTO Account VALUES (102, 'Priya', 15000);
```

```
COMMIT;
```

```
START TRANSACTION;
```

```
UPDATE Account  
SET Balance = Balance - 2000  
WHERE Account_ID = 101;
```

```
SAVEPOINT sp1;
```

```
UPDATE Account  
SET Balance = Balance + 5000  
WHERE Account_ID = 102;
```

```
ROLLBACK TO SAVEPOINT sp1;
```



```
COMMIT;
```

```
SELECT * FROM Account;
```

Here, the ₹5000 change is cancelled, but Rahul's ₹2000 change remains.

5. RELEASE SAVEPOINT Program

SQL

```
CREATE DATABASE BankDB5;
```

```
USE BankDB5;
```

```
CREATE TABLE Account (  
    Account_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Balance INT  
);
```

```
INSERT INTO Account VALUES (101, 'Rahul', 10000);
```

```
INSERT INTO Account VALUES (102, 'Priya', 15000);
```

```
COMMIT;
```

```
START TRANSACTION;
```

```
UPDATE Account  
SET Balance = Balance - 2000  
WHERE Account_ID = 101;
```

```
SAVEPOINT sp1;
```

```
UPDATE Account  
SET Balance = Balance + 2000  
WHERE Account_ID = 102;
```

```
RELEASE SAVEPOINT sp1;
```



```
COMMIT;
```

```
SELECT * FROM Account;
```

RELEASE SAVEPOINT removes the savepoint because it is no longer needed.